

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Ravi KapoorStudent ID: 008630031**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

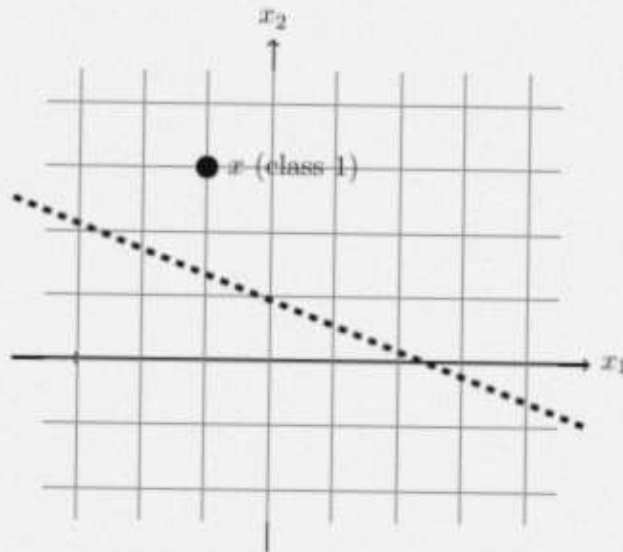
$$\begin{aligned}P(H|K=3) &= P(K=3|H)P(H)/P(K=3) \\P(K=3) &= .4116(.25) + .25(.75) \\&= .1029 + .1875 = .2904 \\&= .1029 / .2904 \\&= 0.354 \\P(H) &= 1/4\end{aligned}$$

Name: Ravi Kapoor

Student ID: 008630031

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{H}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = -3.5/9 \approx -0.39$$

$$x_2 = -0.39 x_1 + 0.99$$

$$w = [0.39, 1]$$

$$b = -0.99$$

$$w \cdot x + b = -0.39 + 3 - 0.99 > 0$$

class 1 ok

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Elena PetrovaStudent ID: 003844862**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

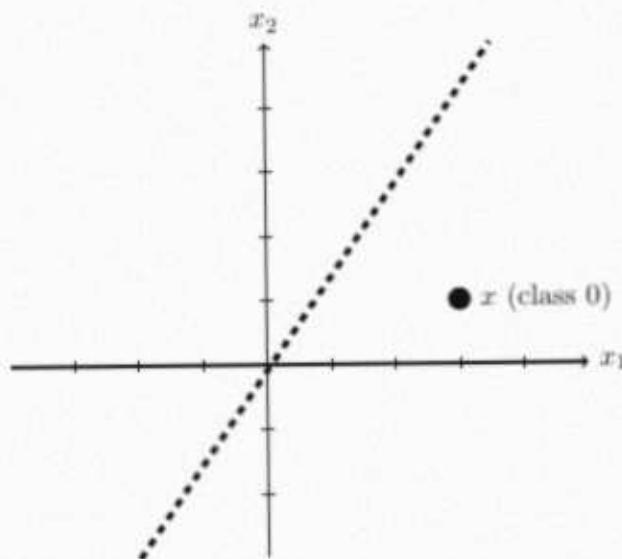
k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned} P(K=2) &= .384(.4) + .375(.6) \\ &= .1536 + .225 = .3786 \\ .1536 / .3786 \end{aligned}$$

Name: Elena PetrovaStudent ID: 003844862**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = 8/5.5 \sim 1.45$$

$$x_2 = 1.45 x_1 - 0.1$$

$$w = [1.45, -1]$$

$$w \cdot x + b = 4.35 - 1 - .1 > 0 ??$$

scratch

$w = [2, -3]$?

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Asha MenonStudent ID: 007126885**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$= .1536 + .225 = .3786$$

$$.1536 / .3786$$

$$P(L) = 0.4$$

scratch

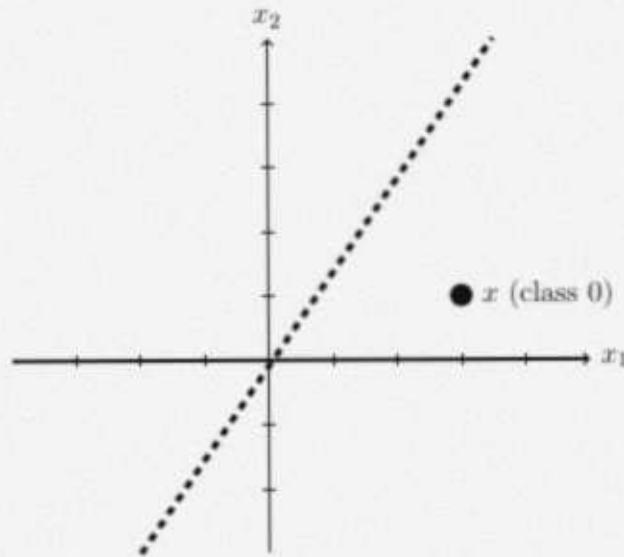
redo arithmetic

Name: Asha Menon

Student ID: 007126885

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}\{w^\top x + b > 0\}$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = 8/5.5 \sim 1.45$$

$$w = [1.45, -1]$$

$$b = -0.1$$

$$w = [2, -3]?$$

scratch

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Hana NovakStudent ID: 007085503**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L \mid K=2) = P(K=2 \mid L) P(L) / P(K=2)$$

$$P(K=2) = .384(.4) + .375(.6)$$

$$= .1536 + .225 = .3786$$

$$= 0.406$$

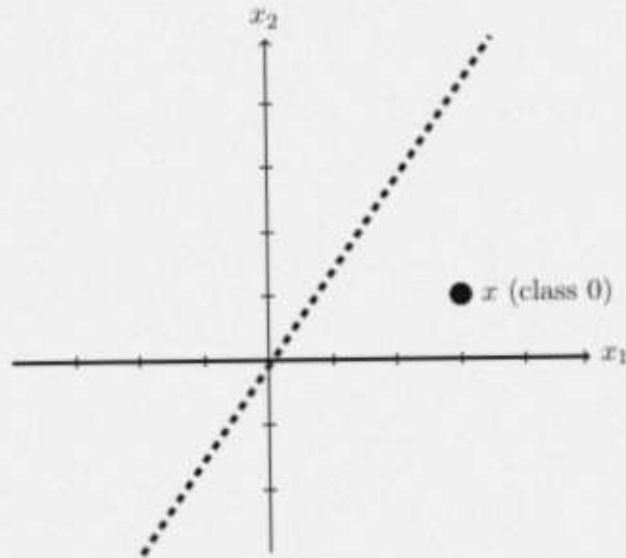
$$\cancel{P(L)} = 0.4$$

Name: Hana Novak

Student ID: 007085503

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = 8/5.5 = 1.45$$

$$x_2 = 1.45 x_1 - 0.1$$

$$w = [1.45, -1]$$

$$b = -0.1$$

$$w \cdot x + b = 4.35 - 1 - 0.1 > 0 \text{ ?}$$

no - any $c \cdot w, c > 0$ works

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Diego RamosStudent ID: 005871000**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

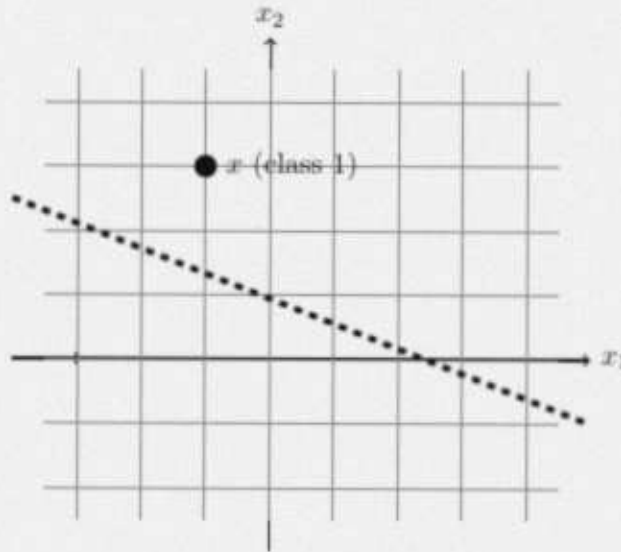
k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}P(K=3) &= .4116(.25) + .25(.75) \\&= .1029 + .1875 \\&= .2904\end{aligned}$$

Name: Diego RamosStudent ID: 005871000**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = -3.5/9 \sim -0.39$$

$$x_2 = -0.39 x_1 + 0.94$$

$$w = [0.39, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -0.39 + 3 - 0.94 > 0$$

class 1 ok

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Jordan BlakeStudent ID: 002629104**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$= .1029 + .1875 = .2904$$

$$.1029 / .2904$$

$$P(H) = 1/4$$

Quiz 1

Probability and linear classifiers
(fictional course, for the `scan_split` example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Ines DuarteStudent ID: 008370680**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2) = .384(.4) + .375(.6)$$

$$.1536 / .3786$$

$$= 0.406$$

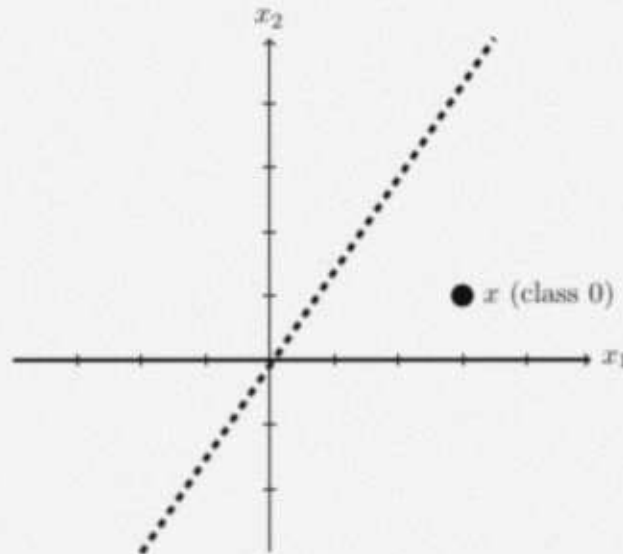
$$P(L) = 0.4$$

Name: Ines Duarte

Student ID: 008370680

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{K}\{w^\top x + b > 0\}$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = \frac{1}{1.45} \approx 0.69$$

$$x_2 = 1.45 x_1 - 0.1$$

$$w = [1.45, -1]$$

$$b = -0.1$$

$$w \cdot x + b = 4.35 - 1 - 0.1 > 0 ??$$

$$\text{flip sign: } w = [-1.45, 1]$$

no - any $c \cdot w, c > 0$ works

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Taylor Brooks

Student ID: 004476855

Problem 1 [10 pts] Which coin?

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

~~$$P(H|K=3) = \frac{P(K=3|H)P(H)}{P(K=3)}$$~~

$$P(K=3) = .4116(.25) + .25(.75)$$

$$= .1029 + .1875 = .2904$$

$$.1029 / .2904$$

$$= 0.354$$

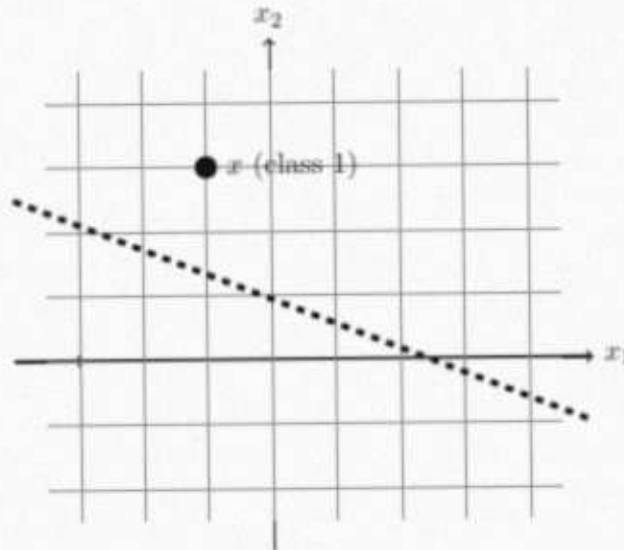
$$P(H) = 1/4$$

Name: Taylor Brooks

Student ID: 004476855

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = -3.5/9 \sim -0.39$$

$$w = [0.39, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -0.39 + 3 - 0.94 > 0$$

redo arithmetic
scratch

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Sofia LindqvistStudent ID: 008112742**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}P(L|K=2) &= P(K=2|L) P(L) / P(K=2) \\&= 0.406 \\P(L) &= 0.4\end{aligned}$$

redo arithmetic

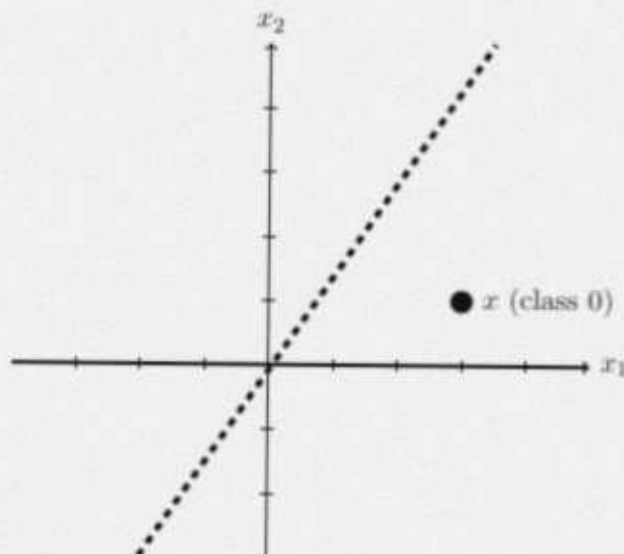
scratch

(cont.)

$w = [2, -3]$?

Name: Sofia LindqvistStudent ID: 008112742**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = 8/5.5 \approx 1.45$$

$$x_2 = 1.45 \times 1 - 0.1$$

$$w = [1.45, -1]$$

$$b = -0.1$$

$$w \cdot x + b = 4.35 - 1 - 0.1 > 0 \text{ ?}$$

$$\text{flip sign: } w = [-1.45, 1]$$

no - any $c \cdot w$, $c > 0$ works

redo arithmetic

~~check: 7556/5786~~

$w = [2, -3]?$

(cont.)

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Omar Sutherland

Student ID: 004488265

Problem 1 [10 pts] Which coin?

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$= .1029 + .1875 = .2904$$

$$.1029 / .2904$$

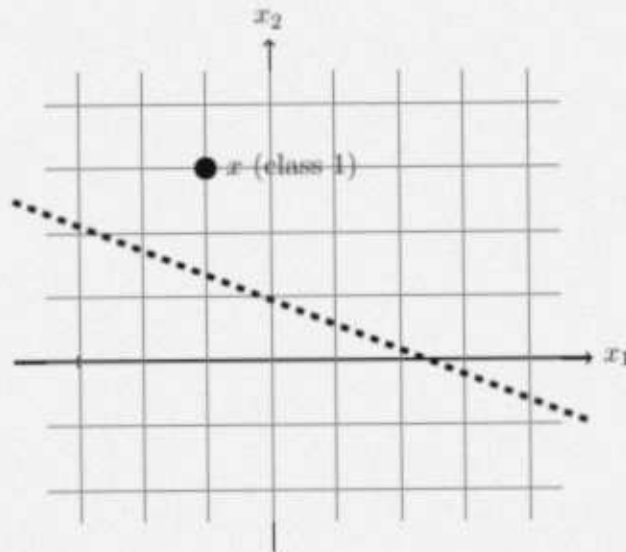
$$P(H) = 1/4$$

Name: Omar Sutherland

Student ID: 004488265

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$w = [0.39, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -0.39 + 3 - 0.94 > 0$$

class 1 ok

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Jonah PierceStudent ID: 00290677**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H \mid K=3) = P(K=3 \mid H) P(H) / P(K=3)$$

$$= .1029 + .1875 = .2904$$

$$.1029 / .2904$$

$$= 0.354$$

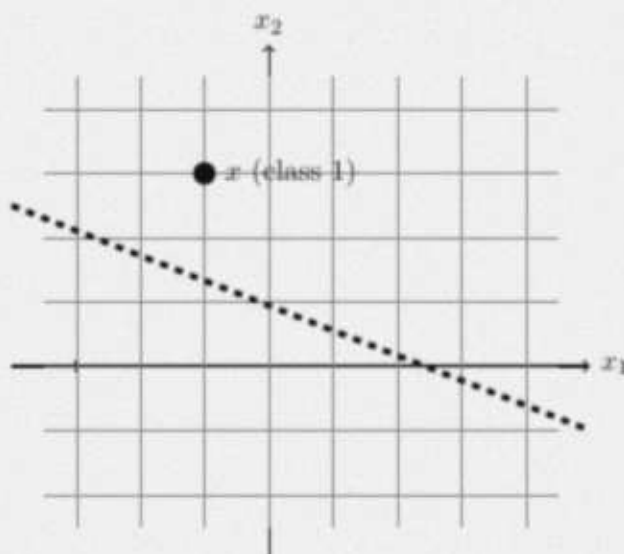
$$P(H) = 1/4$$

Name: Jonah Pierce

Student ID: 002906197

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

~~$$x_2 = 0.39x_1 + 0.94$$~~

$$w = [0.39, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -0.39 + 3 - 0.94 > 0$$

class 1 ok

not unique: scale by $c > 0$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Theo Castillo

Student ID: 001995552

Problem 1 [10 pts] Which coin?

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

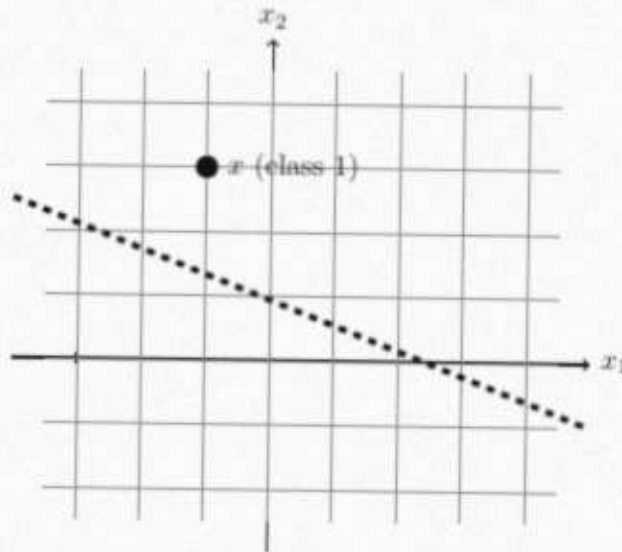
$$\begin{aligned}
 P(K=3) &= .4116(.25) + .25(.75) \\
 &= 0.354 \\
 P(H) &= 1/4
 \end{aligned}$$

Name: Theo Castillo

Student ID: 00199552

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = -3.5/9 \approx -0.39$$

$$x_2 = -0.39 x_1 + 0.94$$

$$w = [0.39, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -0.39 + 3 - 0.94 > 0$$

class 1 ok

not unique: scale by $c > 0$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Lena HoffmannStudent ID: 002043697**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$= 0.1536 / 0.3786$$

$$P(L) = 0.4$$

check: $.1536 / .3786$

$w = [2, -3]^T$?

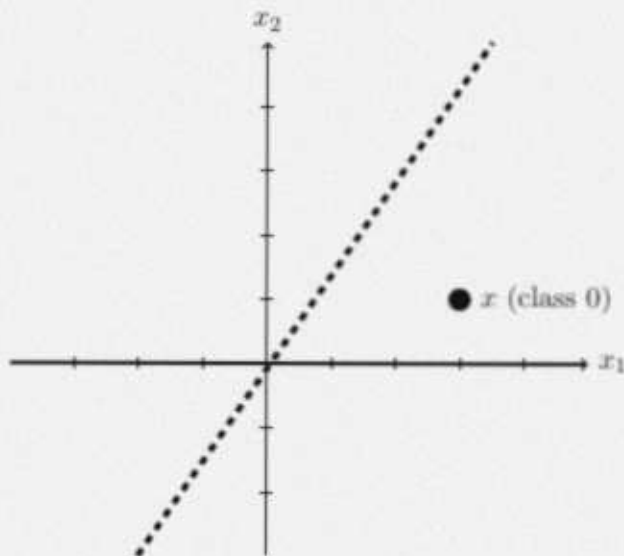
scratch

Name: Lena Hoffmann

Student ID: 002043697

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{K}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$w = [1.45, -1]$$

$$w \cdot x + b = 4.35 - 1 - .1 > 0 ??$$

$$\text{Flip sign: } w = [-1.45, 1]$$

no - any $c \cdot w, c > 0$ works

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Kai YamamotoStudent ID: 001927287**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

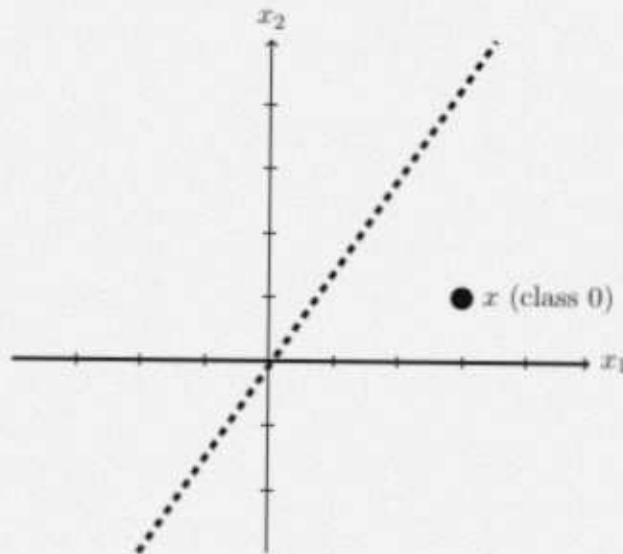
$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$= 0.406$$

$$P(L) = 0.4$$

Name: Kai YamamotoStudent ID: 001927287**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$w = [1.45, -1]$$

$$b = -0.1$$

$$w \cdot x + b = 4.35 - 1 - .1 > 0 ??$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Mina HaddadStudent ID: 003829817**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2) = .384(.4) + .375(.6)$$

$$.1536 / .3786$$

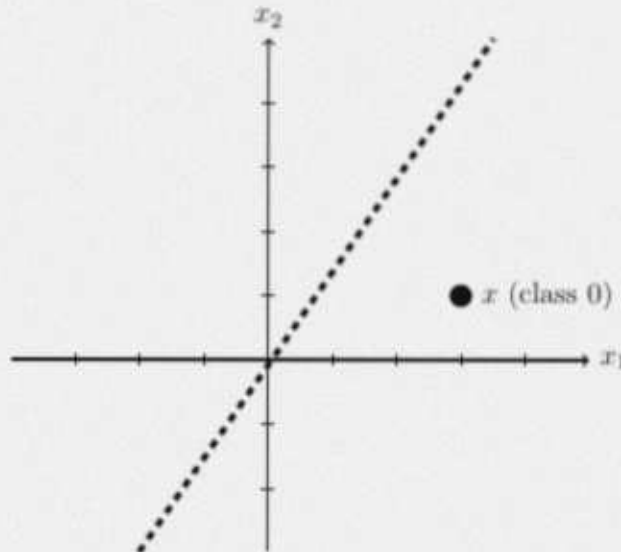
$$= 0.406$$

Name: Mina Haddad

Student ID: 003829817

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = 8/5.5 \approx 1.45$$

$$x_2 = 1.45x_1 - 0.1$$

$$wx + b = 4.35 - 1 - .1 > 0 ??$$

no - any $c^*w, c > 0$ works

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Avery ChenStudent ID: 003510551**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

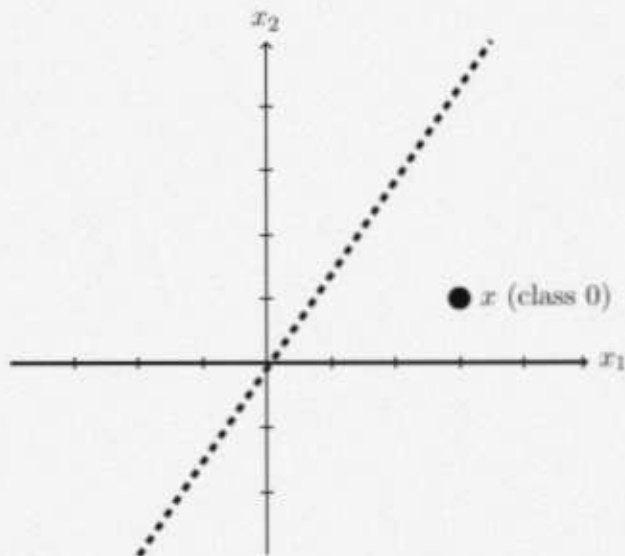
$$P(K=2) = .384(.4) + .375(.6)$$

$$.1536 + .225$$

$$= 0.406$$

Name: Avery ChenStudent ID: 003510551**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = 8/5.5 \approx 1.45$$

$$x_2 = 1.45 x_1 - 0.1$$

$$w = [1.45, -1]$$

$$b = -0.1$$

$$w \cdot x + b = 4.35 - 1 - .1 > 0 ??$$

no - any $c \cdot w$, $c > 0$ works

$$w = [2, -3]?$$

redo arithmetic

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Marcus WebbStudent ID: 001686060**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H)P(H) / P(K=3)$$

$$= .1029 + .1875 = .2904$$

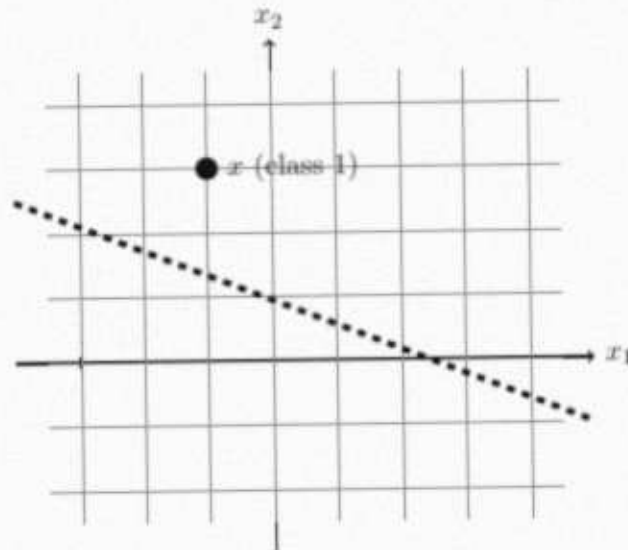
$$.1029 / .2904$$

Name: Marcus Webb

Student ID: 001636060

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}\{w^T x + b > 0\}$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = -3.5/9 = -0.39$$

$$x_2 = -0.39x_1 + 0.94$$

$$w = [0.39, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -3.9 + 3 - 0.94 > 0$$

class 1 ok

not unique: scale by $c > 0$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Priya NatarajanStudent ID: 009232036**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2) = .384(.4) + .375(.6)$$

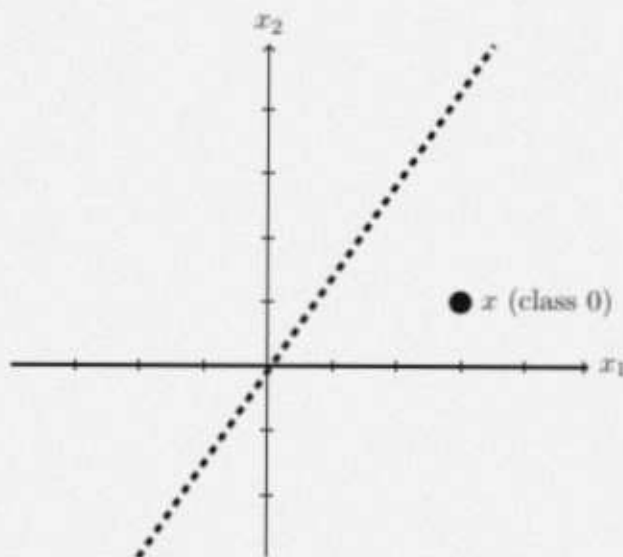
$$= .1536 + .225 = .3786$$

$$.1536 / .3786$$

$$= 0.406$$

Name: Priya NatarajanStudent ID: 009232036**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{H}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = 8/5.5 \sim 1.45$$

$$x_2 \approx 1.45 x_1 - 0.1$$

$$w = [1.45, -1]$$

$$b = -0.1$$

$$\text{flip sign: } w = [-1.45, 1]$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Riley MorenoStudent ID: 005178510**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(K=2) = .384(.4) + .375(.6)$$

$$= .1536 + .225 = .3786$$

$$.1536 / .3786$$

$$= 0.406$$

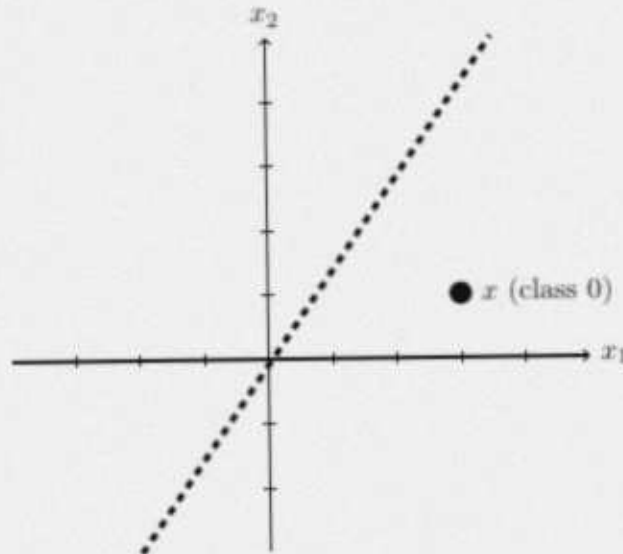
$$P(L) = 0.4$$

Name: Riley Moreno

Student ID: 005178510

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{K}\{w^\top x + b > 0\}$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$x_2 = 1.45 x_1 - 0.1$$

$$w = [1.45, -1]$$

$$\underline{b = -0.1}$$

$$w \cdot x + b = 4.35 - 1 - .1 > 0 ??$$

$$\text{flip sign: } w = [-1.45, 1]$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Felix AdeyemiStudent ID: 006392750**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3) = .4116(.25) + .25(.75)$$

$$= .1029 + .1875 = .2904$$

$$.1029 / .2904$$

$$= 0.354$$

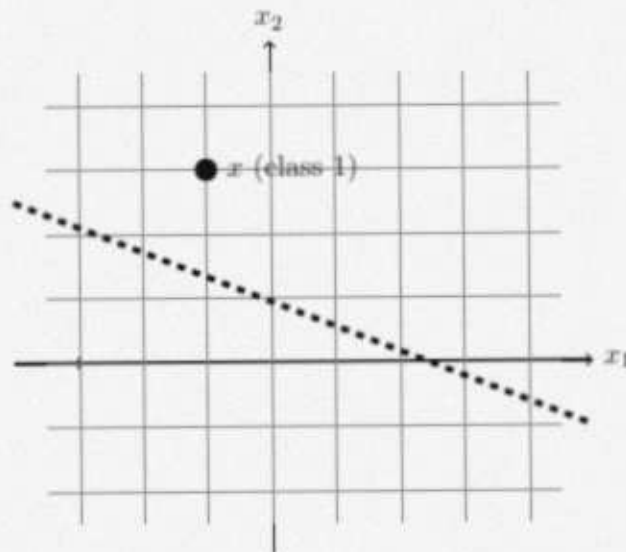
$$P(H) = 1/4$$

Name: Felix Adeyemi

Student ID: 006392750

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{H}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$x_2 = -0.39 x_1 + 0.94$$

$$w = [0.39, 1]$$

$$b = -0.94$$

class 1 ok

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Grace KimStudent ID: 001596831**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2) = .384(.4) + .375(.6)$$

~~$$= .1536 + .225 = .3786$$~~

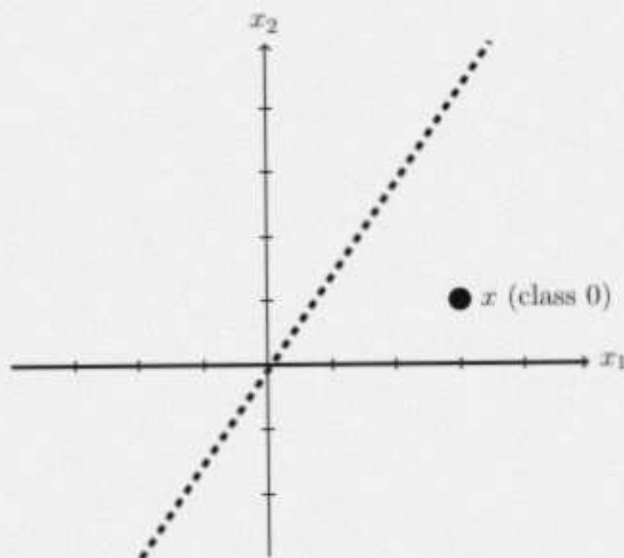
$$.1536 / .3786$$

$$= 0.406$$

$$P(L) = 0.4$$

Name: Grace KimStudent ID: 001596831**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}\{w^\top x + b > 0\}$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$x_2 = 1.45 x_1 - 0.1$$

$$w = [1.45, -1]$$

$$b = -0.1$$

$$w \cdot x + b = 4.35 - 1 - .1 > 0 ??$$

$$\text{flip sign: } w = [-1.45, 1]$$

no - any $c \cdot w$, $c > 0$ works

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Sam OkaforStudent ID: 003051404**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}
 \cancel{P(K=3)} &= \cancel{.4116}(.25) + \cancel{.25}(.75) \\
 .1029 / .2904 \\
 &= 0.354
 \end{aligned}$$

$$P(H) = 1/4$$

(cont.)

check: $.1536/.3786$

scratch

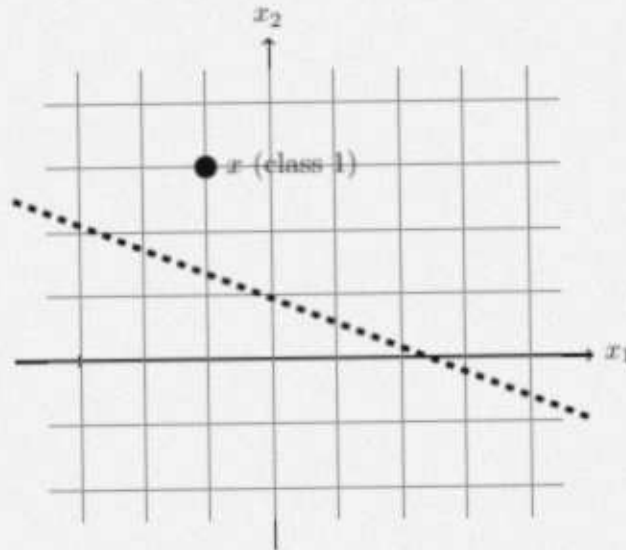
redo arithmetic

Name: Sam Okafor

Student ID: 003051404

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}\{w^\top x + b > 0\}$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = -3.5/1 = -0.37$$

$$x_2 = -0.37x_1 + 0.94$$

$$w = [0.37, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -0.37 + 3 - 0.94 > 0$$

class 1 ok

not unique: scale by $c > 0$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Noor Rahman

Student ID: 009879200

Problem 1 [10 pts] Which coin?

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3) = .4116(.25) + .25(.75)$$

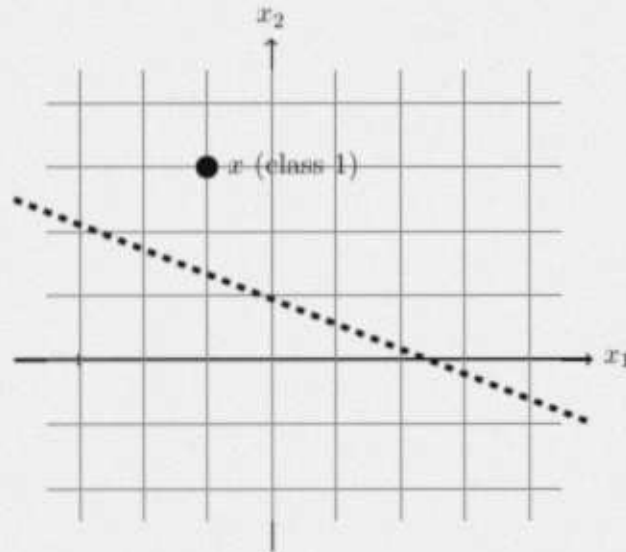
$$= .1029 + .1875 = .2904$$

Name: Noor Rahman

Student ID: 009879200

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = -3.57 - -0.39$$

$$x_2 = -0.39 x_1 + 0.94$$

$$w = [0.39, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -0.39 + 3 - .94 > 0$$

class 1 ok

not unique: scale by $c > 0$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Name: Leo FischerStudent ID: 005914170**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3) = .4116(.25) + .25(.75)$$

$$= .1029 + .1875 = .2904$$

$$.1029 / .2904$$

$$= 0.354$$

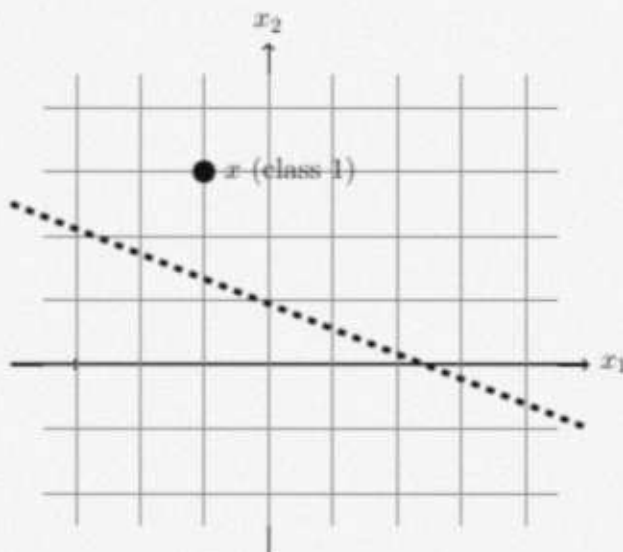
$$P(H) = 1/4$$

Name: Leo Fischer

Student ID: 005914170

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = -3.5/9 = -0.39$$

$$x_2 = -0.39 x_1 + 0.94$$

$$w = [0.39, 1]$$

$$b = -0.94$$

$$w \cdot x + b = -3.9 + 3 - 0.94 > 0$$

class 1 ok

not unique: scale by $c > 0$

